

Closing the Pre-Infusion Gap: A Quality Improvement Initiative to Enhance Laboratory Readiness for Outpatient IV Iron Referrals

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Laboratory completion increased from 69.1% to 89.8%

Laboratory treatment-parameter attainment remained high: 90.8% at baseline and 94.9% after PDSA 2.

MOTIVATION

At Renown in Northern Nevada, some patients arrived for IV iron infusions without valid iron studies, requiring same-day testing and coordination among nursing staff, pharmacy, and referring clinicians. Lab orders embedded in treatment plans did not reliably ensure readiness before arrival, highlighting the need for earlier verification.

★ **Nearly 1 in 3 lacked valid iron studies before infusion**
Baseline: 64/207 | Jan–Apr 2026

AIM

Increase required iron study completion to ≥85% within six months among internal IV iron referrals, supported by verification before scheduling.

ESTIMATED CHAIR-TIME BURDEN

Baseline period: Jan–Apr 2026
Assuming 1 additional chair-hour per affected visit

16 additional chair-hours/month
≈\$18,600/month in rate-based value
Annualized: 192 hours ≈ \$223,000*

**Estimates use a \$1,161.30/hour rate proxy and assume one additional chair-hour per affected visit. Annual projections assume baseline volume and sustained period-specific rates; actual delays, costs, and savings were not measured.*

METHODS

Single-center QI evaluation of 416 internal outpatient IV iron referral records, January 1–September 14, 2026. All records were included regardless of treatment status.

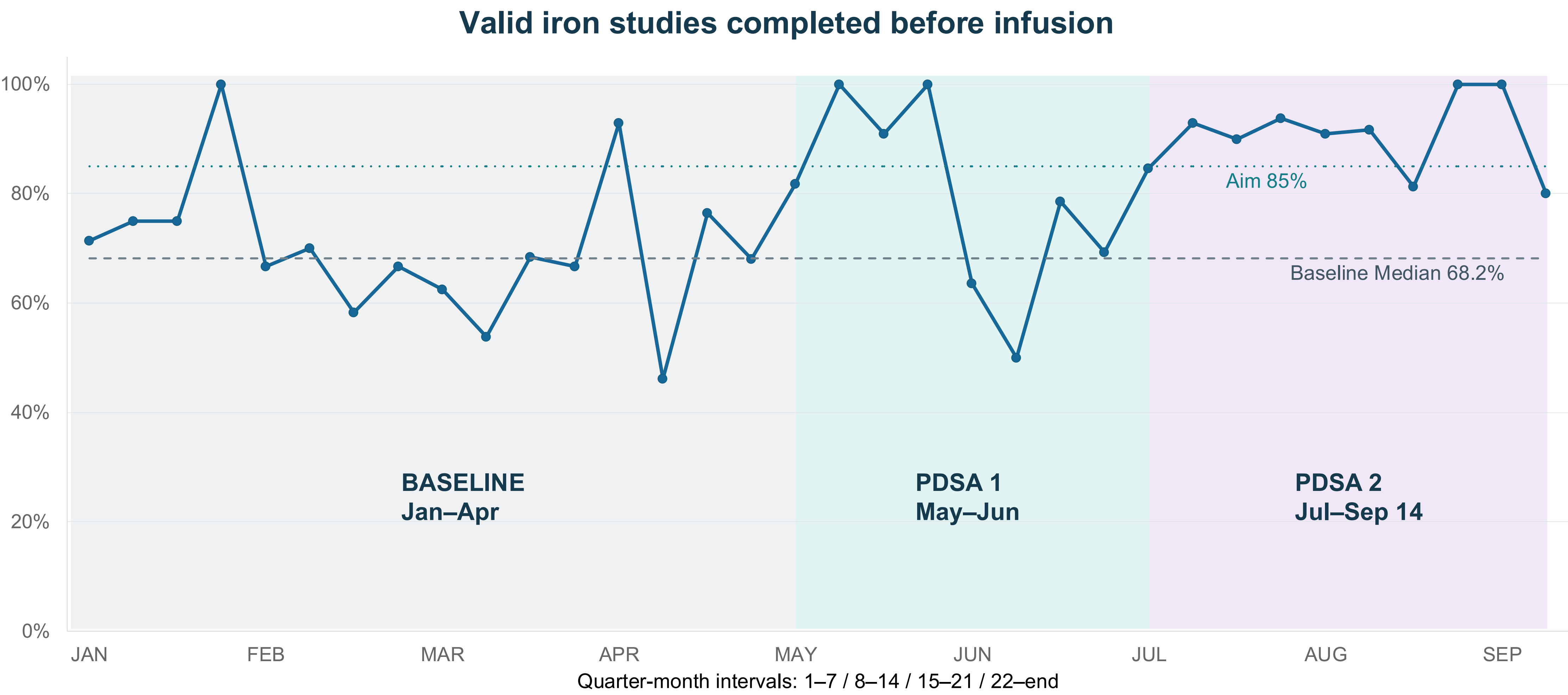
Process measure: Percentage with complete, valid iron studies before infusion.

Outcome measure: Percentage meeting laboratory treatment parameters.

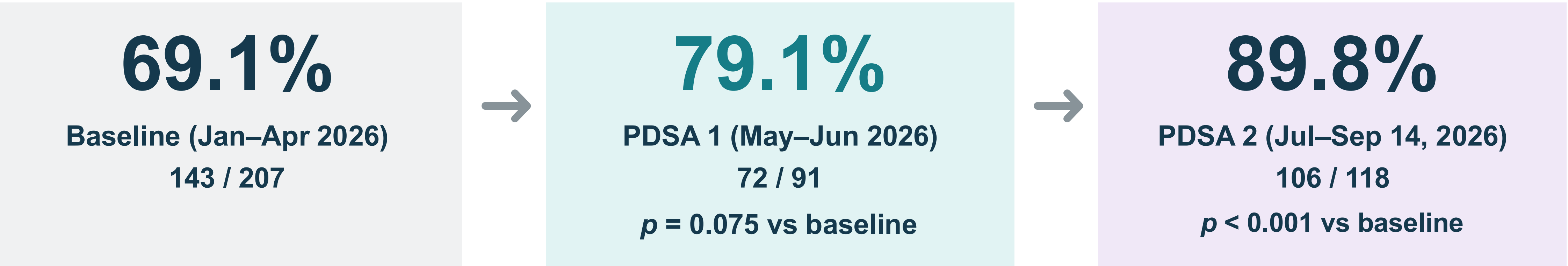
Analysis:

- Quarter-month run chart displayed process performance over time using a fixed baseline median.
- Pearson’s chi-square test compared pooled laboratory readiness rates across the three study periods: baseline, PDSA 1, and PDSA 2, assuming independent records. Outcome rates were summarized descriptively.

PROCESS MEASURE: LABORATORY READINESS OVER TIME

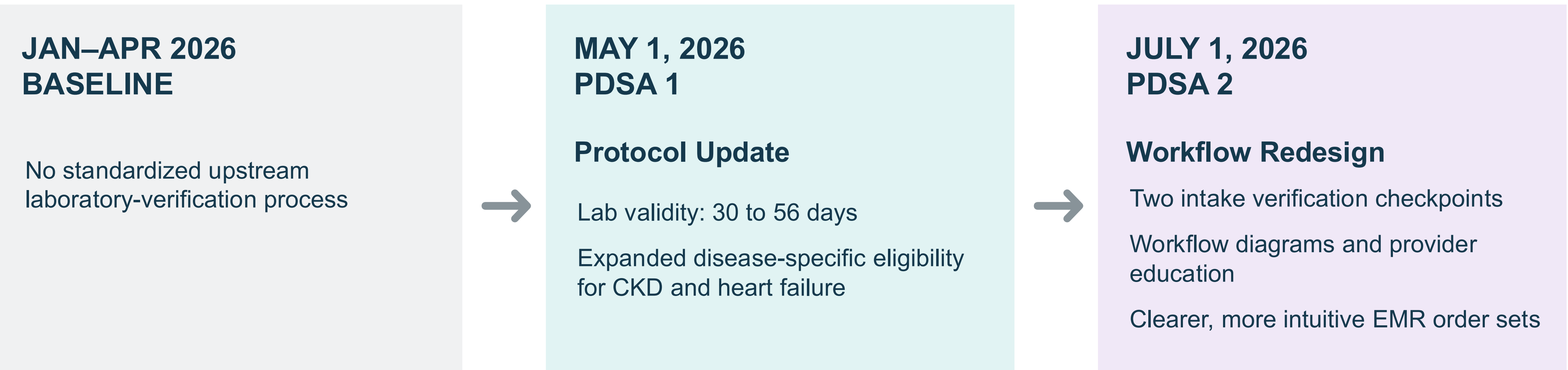


LABORATORY READINESS BY PERIOD



Overall $p < 0.001$ across three periods. Pairwise p-values compare each intervention period with baseline using Pearson’s chi-square tests with Holm adjustment.

WHAT WE CHANGED



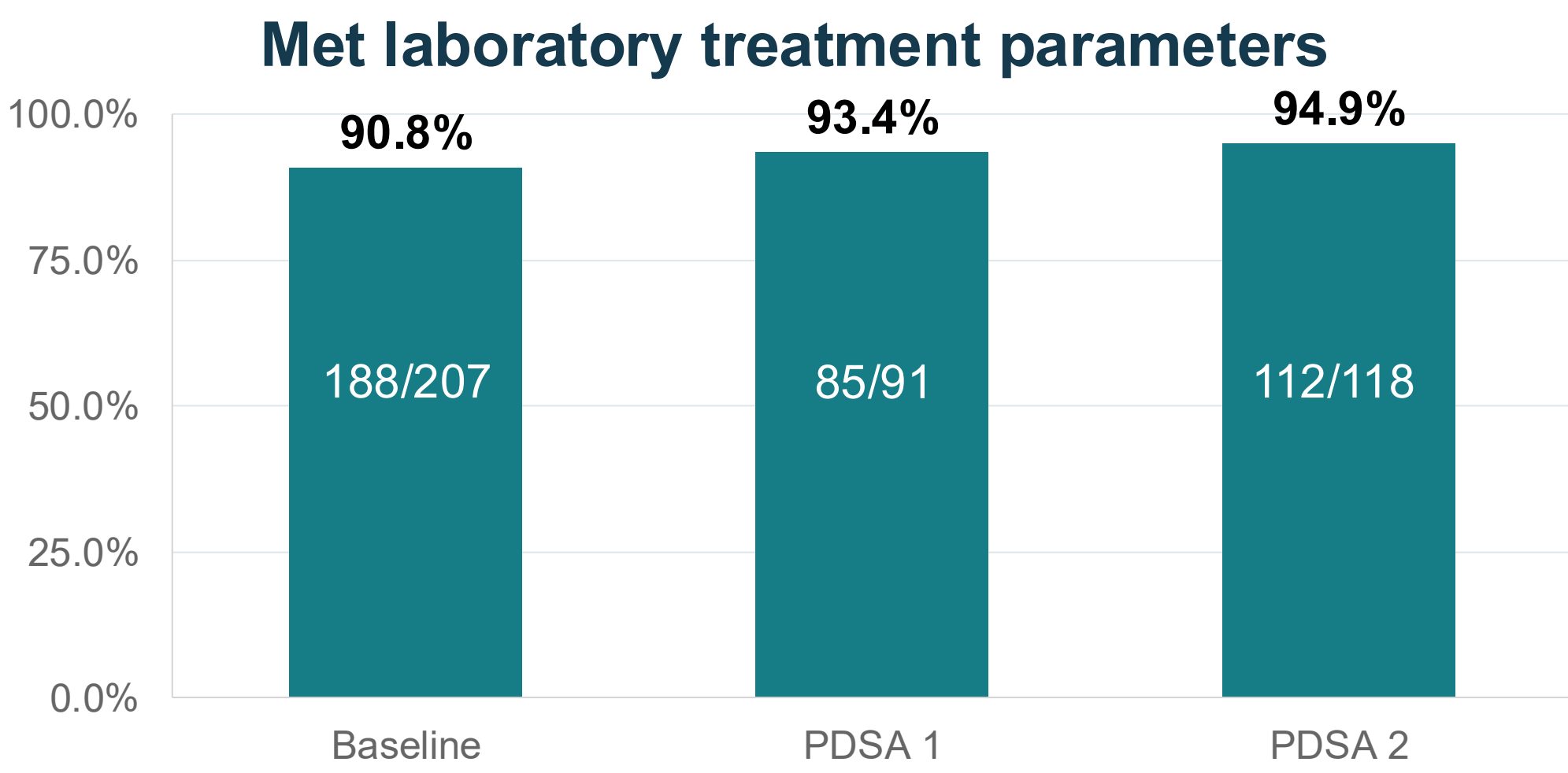
ACKNOWLEDGMENTS

We thank Renown’s infusion nursing, pharmacy, and intake teams for implementation support, and American Society of Hematology mentors Drs. Ming Lim and Richard Godby for their guidance.

REFERENCES

Renown Health. *Pricing Transparency*. Accessed September 30, 2026. <https://www.renown.org/Patients-and-Visitors/Billing/Pricing-Transparency>

OUTCOME MEASURE



Attainment remained high across all periods.

PATIENT IMPACT (ESTIMATED)

≈129 patient-hours of extra chair time potentially avoided annually*

Potentially less waiting, fewer treatment disruptions, and less uncertainty for patients and caregivers, including those traveling from rural Northern Nevada.

**Uses the same chair-time model as the capacity estimates below, assuming baseline volume, one extra hour per affected visit, and sustained improvement. Patient benefits were not directly measured.*

MODELED ANNUAL CAPACITY BURDEN

Modeled annual burden*	Baseline	PDSA 1	PDSA 2
Additional chair-hours	192	130	63
Rate-based value	\$223,000	\$151,000	\$73,000

** Annualized at baseline volume (621 visits/year), assuming one additional chair-hour per visit lacking valid studies and \$1,161.30/hour. Projections assume sustained period-specific rates; not measured delays or financial savings.*

LIMITATIONS

Single-center, uncontrolled evaluation with limited follow-up. Changes in laboratory validity and eligibility criteria may affect comparisons across periods. Individual intervention effects cannot be isolated. Chair-time and financial estimates are modeled, not directly measured.

CONCLUSION

Protocol and workflow changes were associated with improved laboratory readiness, exceeding the 85% target overall after PDSA 2. Laboratory treatment-parameter attainment remained high. Intermittent below-target intervals highlight the need for continued monitoring and refinement.

NEXT STEPS

Monitor sustainability and adapt the workflow for an external-referral pilot.

